
Machine Learning – 2026

COVID-19 Dataset Prediction Project

Deadline: July 20, 2026, 23:55

Submission portal: ILIAS

Required files: one PDF report and one completed notebook, named `project_notebook.ipynb`

Team size: individual or groups of up to 2 students

Report length: at most 8 pages, excluding references; appendices are not graded. Conciseness is valued: clear and focused writing is preferred over exhaustive coverage.

Overview

In this project you will search for a dataset published about COVID-19, choose a supervised classification or regression task, and document your full modeling journey. Classification examples include predicting whether a patient will be hospitalized, survive, or require intensive care. Regression examples include predicting hospital-stay duration or a future case count. You can choose a different supervised prediction task, as long as it is supported by the dataset you select and relevant for understanding the impact of COVID-19.

Your first task is to select a suitable dataset and define a feasible prediction task. You may use the dataset suggested below or choose another. In your report, explain what the dataset represents, justify its suitability for your task, and discuss how limitations in its data-collection process affect your modeling decisions and the interpretation and generalizability of your results. You are not expected to find a perfectly unbiased dataset.

Your goal is not only to obtain the highest possible predictive score. You should show how you made modeling decisions, what you learned from the data, which preprocessing choices helped, and how your final model behaves under a careful evaluation protocol. A well-executed study that finds little predictive signal can receive full credit if the methodology and interpretation are sound.

At the beginning of your report, explain whether the data-collection process permits your selected target to be interpreted as a population-level outcome. Distinguish the population represented in the dataset from the population to which you want to generalize.

Support

You are welcome to ask for guidance and help during the exercise and tutorial sessions throughout the semester.

Suggested Dataset

One suggestion is the [Mexico COVID-19 Clinical Data dataset](#), based on records published by the Government of Mexico through SISVER, the epidemiological surveillance system of Mexico's Ministry of Health for viral respiratory diseases. A natural prediction task on this dataset is to predict whether a patient is hospitalized.

You are free to use this dataset or to choose another COVID-19 dataset. If you choose another dataset, explain why it is appropriate, informative, or interesting for your selected prediction task.

According to the dataset description:

- 475 hospitals and clinics were selected as sentinel reporting units across the country.
- Every patient presenting with respiratory symptoms was evaluated using a clinical case definition.
- Testing was rationed by severity: 100% of patients with severe symptoms or SARI (Severe Acute Respiratory Infection) were tested.
- Only 10% of patients with mild symptoms were tested, using sampling rather than universal testing.
- Asymptomatic individuals were essentially never tested.

Note that several variables in this dataset (e.g., PNEUMONIA, INTUBED, ICU) are typically recorded at *discharge*, not at the time of initial presentation. Treat them as potential leakage: they may reflect the outcome of the hospital stay rather than information available at prediction time. Justify carefully any such variable you choose to include.

Required Report Structure

Your report must contain the following three sections.

A. Problem Formulation and Dataset Analysis

Explain why you chose your dataset and define your prediction task precisely. State whether it is classification or regression, and specify the target, prediction population, prediction moment, prediction horizon, and information available at prediction time. Describe the input variables and high level statistics such as the number of observations, missing values, target or class distribution, and important caveats in the data collection process.

Include a short leakage audit for important or potentially problematic variables. For example:

Variable	Available at prediction time?	Included?	Reason
Age	Yes	Yes	Baseline information
ICU status	No	No	Downstream of admission

Explore the data visually and extract as much useful information as you can. Include at least three meaningful plots. Good plots should help you understand the prediction task, reveal potential confounders, expose data quality issues, or motivate preprocessing choices.

Include a brief reflection on fairness and potential harms. Consider which subgroups are over- or under-represented in the data, whether the model could perform systematically worse for some groups (e.g., by age, sex, or socioeconomic access to care), and who would be affected by errors in deployment. This reflection does not need to be long, but it must be substantive.

Evaluate at least one meaningful preprocessing or feature-design choice through an ablation or controlled comparison. Explain what you compared, what you observed, and why you kept or rejected the choice.

B. Model Comparison

Compare three genuinely different model families appropriate to your task. Include one simple, interpretable model, such as logistic regression for classification or linear regression for regression, and at least two other appropriate model families.

Before running the experiments, write which method you intuitively expect to work best and why. Then compare the methods empirically using appropriate metrics for your task. Compare them against a trivial reference predictor, such as a majority-class or training-prevalence predictor for classification, or a training-mean or training-median predictor for regression. If the result is different from your expectation, explain your intuition for why this happened. Consider factors such as target distribution, class imbalance where applicable, nonlinear relationships, feature interactions, overfitting, data leakage, missing values, and dataset bias.

C. Final Pipeline, Cross-Validation, and Hyperparameter Search

For your final model, wrap the entire preprocessing and prediction workflow into a pipeline or custom class. The block should include all transformations that are learned from data, such as imputation, scaling, encoding, feature selection, and the predictive model itself. The test set must not be used to fit preprocessing steps or to choose hyperparameters.

Choose a train/test split and cross-validation strategy appropriate to the dataset. For example, temporal, grouped, or repeated-patient data may require a non-random split. Explain and justify your choice.

Run cross-validation on the training data using the whole pipeline. Use this cross-validation to perform hyperparameter selection for your final model. Brute-force search and large compute budgets are not rewarded; aim for at most around 200 total fitted models across your entire hyperparameter search. Report:

- the cross-validation protocol you used;
- the rationale for your train/test split;
- the hyperparameters you searched over;
- the number of candidate configurations and total fitted models;
- the best hyperparameters selected;
- the validation performance during model selection;
- the final performance on the held-out test set.

Analyse where your final model fails. Examine which subgroups, feature ranges, or input patterns are associated with the highest errors or most frequent misclassifications. Do the failure patterns suggest a systematic bias, a data quality issue, or a genuine limit of predictability? A brief but honest error analysis is more valuable than simply reporting aggregate metrics.

Evaluation Guidance

Choose metrics that match your dataset, target, and intended use. For classification, accuracy alone may be misleading for imbalanced outcomes; consider precision, recall, F1 score, ROC-AUC, PR-AUC, confusion matrices, or probability calibration where appropriate. For regression, consider MAE, RMSE, R^2 , residual plots, and task-relevant error tolerances. Report at least one primary metric and justify why it reflects the costs and goals of your prediction task.

When working with patient-level data, be especially careful about leakage. Do not include features that are direct consequences of the outcome if the prediction task is supposed to happen before the outcome is known.

Marking Rubric

The project is marked out of 30 points. Predictive performance alone does not determine the grade.

Component	Points
Problem formulation and dataset criticism	3
Exploratory analysis and preprocessing evidence	3
Validation design and leakage prevention	6
Model comparison and experimental reasoning	6
Final pipeline, evaluation, and error analysis	4
Reproducibility and code quality	2
Report clarity and citation	1
Oral presentation	5
Total	30

Deliverables

Submit the following files to ILIAS by July 20, 2026, 23:55:

- A PDF report containing the three required sections, figures, tables, and explanations.
- The completed notebook `project_notebook.ipynb` with all cells executed in order.
- Clear instructions for obtaining the dataset if it is not included directly.

Oral Presentation

All teams are required to present their project during the final week of the lecture, on July 21 or July 22, 2026. Further details (schedule, format, and duration) will be announced separately.

Collaboration and Citation Policy

Students may discuss high-level ideas, but each submitted report and notebook must be written by the submitting team/person. Cite all resources used, including datasets, papers, documentation, online discussions, previous course material, and AI assistants. When using AI tools, briefly state how they were used, for example for debugging, rephrasing, brainstorming, or checking an explanation.

Beyond the mandatory presentation (see above), we may ask individual authors to discuss specific details of their code and decisions for academic-integrity verification. These checks are not a separately graded rubric component, but inability to explain submitted work may trigger the applicable academic-integrity process. If you used AI tools, be prepared to explain the reasoning behind your modeling choices and the content of your report without relying on AI-generated explanations.

Reproducibility Checklist

Before submitting, make sure that:

- The notebook runs from top to bottom without manual edits.
- Random seeds are fixed where appropriate.
- Figures and tables in the report match the executed notebook.
- Dataset sources and access dates are documented.
- Package versions or environment setup steps are recorded.
- Train/test indices are created and saved before model selection.
- Test labels are not examined during model selection.
- The final test-set result is produced only after model selection.